

CSE 5825 — Homework 1 Study Notes

Tools You Need, Not the Answers

These notes collect the *techniques* each problem is testing. No final numeric or symbolic answers to your specific problems are given — the goal is that you can produce them yourself.

Problem 1: Product Rule, Chain Rule, Bayes' Rule

Everything here comes from one definition, applied twice:

$$p(u | v) = \frac{p(u, v)}{p(v)} \iff p(u, v) = p(u | v) p(v).$$

Chain rule with an extra conditioning variable z . The identity still holds if every term is conditioned on the same extra variable z throughout:

$$p(x, y | z) = p(x | y, z) p(y | z) = p(y | x, z) p(x | z).$$

This is just the ordinary product rule applied *inside* the conditional distribution $p(\cdot | z)$ — think of $p(\cdot | z)$ as “a probability distribution in its own right” and apply the same two-variable rule you already know to it. To get the first identity you’re asked to prove, apply the product rule to x and y within the world where z is already given.

Turning it into Bayes’ rule. Once you have $p(x, y | z) = p(x | y, z) p(y | z)$, and separately $p(x, y | z) = p(y | x, z) p(x | z)$, set the two right-hand sides equal to each other and solve for $p(x | y, z)$. That produces the second identity: a version of Bayes’ rule where everything is additionally conditioned on background information z . This “conditioned throughout” form of Bayes’ rule is used constantly in graphical models (e.g. conditioning on evidence you’ve already observed).

Problem 2: Law of Total Probability + Bayes' Rule

This is a two-stage random experiment: first a department is chosen, then a student within it.

- **Part (a)** is the *law of total probability*: to get the overall probability of an event (female), sum the probability of that event *within* each department, weighted by the probability of landing in that department:

$$P(\text{female}) = \sum_{\text{dept}} P(\text{female} | \text{dept}) P(\text{dept}).$$

- **Part (b)** is Bayes’ rule run backwards: you’re given the outcome (female) and asked about the cause (which department).

$$P(\text{dept} = B | \text{female}) = \frac{P(\text{female} | \text{dept} = B) P(\text{dept} = B)}{P(\text{female})}.$$

Note the denominator is exactly the quantity you computed in part (a) — don't recompute it separately.

- **Part (c)** is the same idea as (b) but conditioning on “male” instead of “female.” You'll need $P(\text{male} \mid \text{dept}) = 1 - P(\text{female} \mid \text{dept})$ and the analogous total-probability denominator for male.

A tree diagram (department branches, then male/female sub-branches with the given percentages as edge probabilities) makes all three parts mechanical: multiply along branches to get joint probabilities, sum branches for totals, divide for conditionals.

Problem 3 & 6(d): Counting Parameters of a Discrete Distribution

Full joint table, no structure assumed. If variable i can take k_i values, a full joint table over variables $X_1, \dots, X_m$ has $\prod_i k_i$ entries, but the probabilities must sum to 1, so only $(\prod_i k_i) - 1$ of them are *free* (independent) parameters — the last one is determined by the constraint that everything sums to 1.

A single conditional probability table (CPT), $P(X \mid \text{Parents})$. If X has k values and its parents jointly have r possible configurations, then for *each* of the r parent configurations you need a separate distribution over X 's k values, and each of those distributions has $k - 1$ free parameters (again, one value is determined by the sum-to-1 constraint). So a CPT costs

$$r \times (k - 1)$$

free parameters, where r is the product of the number of values of each parent (or $r = 1$, i.e. just $k - 1$ parameters, if X has no parents).

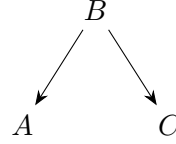
Total parameters for a factorized/network distribution. Add up the CPT cost (using the formula above) for every variable in the factorization. Compare this total to the full-joint-table count from part (a)/(b) of Problem 3 — the gap between them is exactly what structure (conditional independence) buys you.

Problems 3(c), 4(a), 5(a), 6(a)-(b): Reading Independence off a Graph (d-separation)

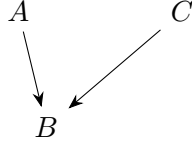
A Bayesian network is a DAG where each node's CPT is $P(\text{node} \mid \text{parents of node})$, and the joint is the product of all these CPTs (this *is* the factorization/chain rule from Problem 1, applied along the graph). Whether two variables are independent (unconditionally, or given some evidence set Z) is read off the graph using three local patterns along any path between them:



Chain: $A \rightarrow B \rightarrow C$
blocked iff B observed



Fork: $A \leftarrow B \rightarrow C$
blocked iff B observed



Collider: $A \rightarrow B \leftarrow C$
blocked iff B (or a
descendant of B) is
not observed

How to use this for $A \perp C \mid Z$: enumerate every undirected path between A and C in the graph. If *every* path is blocked according to the rules above (given the observed set Z), then A and C are independent given Z . If even one path is unblocked, they are (in general) *not* independent given Z . Two things people forget:

- A chain or fork is blocked *only if the middle node is in Z* ; a collider is blocked *only if the middle node (and none of its descendants) is in Z* — this is the opposite condition, which is exactly why colliders create the “explaining away” effect (Problem 4, item 3 vs. item 4 is testing precisely this asymmetry).
- You must check *all* paths, not just the shortest or most obvious one.

Problems 4(b), 5(b)-(c): Computing Marginals/Conditionals from a Factorized Joint

Once you have the joint as a product of CPTs (from the graph structure), any marginal or conditional you want is computed by two mechanical steps:

1. **Marginalize (sum out) variables you don’t want**, e.g.

$$p(d) = \sum_{a,t,s,l,b,e,x} p(a,t,s,l,b,e,x,d).$$

Substitute the chain-rule factorization for the joint, then push each sum as far right as it will go (sum over a variable only after every factor involving it has been multiplied in) — this is the idea behind *variable elimination*, and it’s far less error-prone than expanding the full 2^8 -row table by hand.

2. **Condition by restricting and renormalizing**, e.g. for $p(d \mid s = \text{tr})$: fix $s = \text{tr}$ everywhere in the joint, sum out everything except d , and you already have an unnormalized version of $p(d, s = \text{tr})$; since $p(d \mid s = \text{tr}) = p(d, s = \text{tr})/p(s = \text{tr})$, either divide by $p(s = \text{tr})$ explicitly, or just normalize your two unnormalized values ($d = \text{tr}$ and $d = \text{fa}$) so they sum to 1 — both give the same answer.

For Problem 5(b)-(c), do the same thing symbolically instead of numerically: write $p(x \mid z) = p(x, z)/p(z)$, expand $p(x, z)$ by summing the full joint $p(x, y, w, z)$ over the variables not appearing (y and w), and simplify using the given factorization $p(x, y, w, z) = p(z \mid w)p(w \mid x, y)p(x)p(y)$.

Problem 5(d): Turning “Equal Formulas” into an Independence Condition

$x \perp y \mid z$ means, by definition, $p(x \mid y, z) = p(x \mid z)$ (equivalently $p(x, y \mid z) = p(x \mid z)p(y \mid z)$). Using Bayes’ rule (Problem 1’s identity!) you can also write $p(x \mid y, z) = \frac{p(y \mid x, z)p(x \mid z)}{p(y \mid z)}$.

Comparing this to the definition, $x \perp y \mid z$ holds exactly when $p(y \mid x, z) = p(y \mid z)$ — i.e. knowing x (in addition to z) doesn’t change the distribution of y . Use your answers to parts (b) and (c) (formulas for $p(x \mid z)$ and $p(y \mid z)$ in terms of the given CPTs) to check whether this equality is forced by the given factorization, or only holds under some extra condition — state which.

Practical tips

- For Problem 4(b), you’re explicitly allowed to write code instead of hand-computing 8-variable sums. A skeleton notebook for this is provided separately (`chest_clinic_skeleton.ipynb`) — fill in the CPTs from the table in the assignment and use nested loops / `itertools.product` to enumerate all variable assignments.
- Double check every CPT you write sums to 1 over the child variable for each fixed parent configuration — this is the single most common source of arithmetic errors on these problems.
- For graph problems (3c, 4a, 5a, 6a-b), draw the graph and physically trace each path with your finger while checking the chain/fork/collider rule — don’t try to do it purely symbolically in your head.